JSD

average of 2 KL divergences

lower means more similar

$$JSD(P\|Q) = \frac{1}{2} D_{KL}(P\|M) + \frac{1}{2} D_{KL}(Q\|M)$$

Probability Distributions (that we are comparing)

measures how different P from M.

measures how different Q from M.

KL formula

$$D_{KL}(P\|Q) = \sum_{x} P(x) \cdot \log\left(\frac{P(x)}{Q(x)}\right)$$

- $D_{KL}(P\|Q) \neq D_{KL}(Q\|P)$ — how different Q from P.
- O/p is not distance – it tells you how different two distributions are
- P & Q are same, KL divergence = 0

KL

Given

$P = [0.9, 0.1] \rightarrow$ real

$Q = [0.5, 0.5] \rightarrow$ predicted

$$D_{KL}(P \| Q) = 0.9 \log_2\left(\frac{0.9}{0.5}\right) + 0.1 \cdot \log_2\left(\frac{0.1}{0.5}\right)$$

$$= 0.9(0.847) + 0.1(-2.322)$$

$$= 0.762 - 0.232$$

$$= 0.53 \text{ bits}$$

$\rightarrow$ Q is not great match for P.

JSD

<u>Given</u>

$P = [0.5, 0.3, 0.2] \rightarrow$ Real

$Q = [0.2, 0.2, 0.6] \rightarrow$ predicted / generated

<u>step 1</u> :

find avg distribution M

$$M = \frac{1}{2}(P+Q)$$

$$= \frac{1}{2}\left[\frac{0.5+0.2}{2}, \frac{0.3+0.2}{2}, \frac{0.2+0.6}{2}\right]$$

$$= [0.35, 0.25, 0.4]$$

<u>step 2</u> :

compute KL $D_{KL}(P||M)$

$$D_{KL}(P||M) = \sum P(x) \cdot \ln\left(\frac{P(x)}{M(x)}\right)$$

1) $0.5 \cdot \ln(0.5/0.35) = 0.178$
2) $0.3 \cdot \ln(0.3/0.25) = 0.0546$
3) $0.2 \cdot \ln(0.2/0.4) = -0.1386$

$$D_{KL}(P||M) = 0.094$$

Step 3:

Compute $D_{KL}(Q||M)$

1) $0.2 \times \ln(0.2/0.35) = -0.112$
2) $0.2 \times \ln(0.2/0.25) = -0.0446$
3) $0.6 \times \ln(0.6/0.4) = 0.243$

$$D_{KL}(Q||M) = 0.0864$$

Step 4:

$$= \frac{1}{2}(0.094 + 0.0864)$$

$$= 0.0902.$$

$JSD \approx 0.0902$

— distributions are somewhat different but not totally opposite
— JSD stays betn 0 to 1 so this is a low-to-moderate defference.